

Group P06

Anomalous Login Detection

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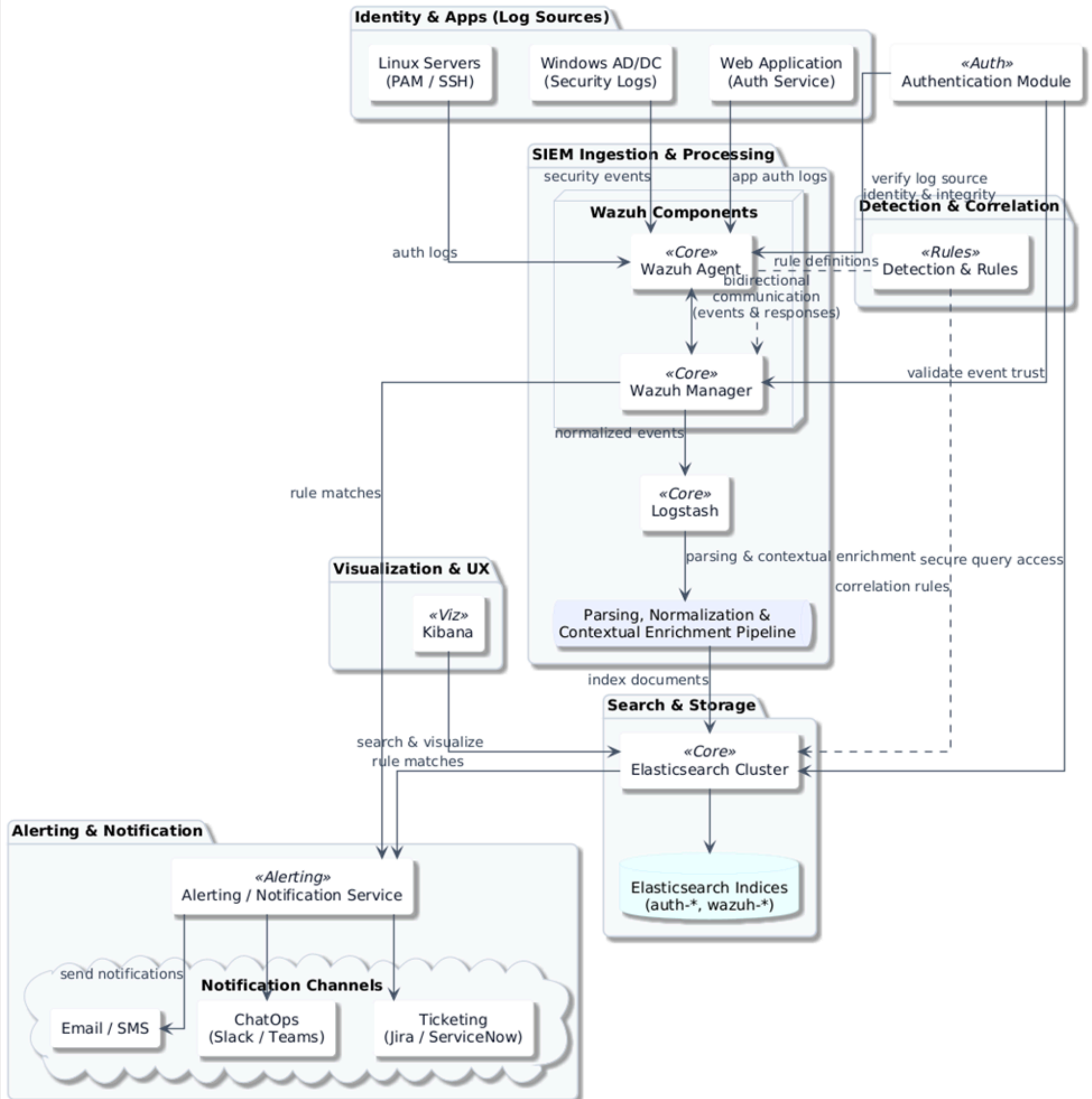
Overview

- **Objective:** Detect suspicious or unusual login activities in real-time using the ELK Stack (Elasticsearch, Logstash, Kibana) integrated with Wazuh.
- **Function:** Collects authentication logs, detects patterns like multiple failed attempts or logins from abnormal locations.
- **Users:** System administrators and security engineers.
- **Key Benefit:**
 - Real-time alerts for abnormal logins.
 - Faster detection and response to security threats.
 - Centralized dashboard for visualization and monitoring.

System Architecture

Suspicious Login Detection - Component Architecture ===

Refined Vertical Layout • Contextual Enrichment • Authentication Layer



System Architecture

- **Core Components:**
- Log Sources (Identity & Apps): Generate authentication/security logs from servers and web apps.
- Wazuh Agent: Collects and forwards logs securely.
- Wazuh Manager + Logstash: Normalize and enrich logs; apply detection rules.
- Elasticsearch Cluster: Stores and indexes events for fast search.
- Kibana: Provides dashboards for analysis and visualization.
- Alerting & Notification System: Sends alerts via Email, SMS, or ChatOps tools.
- **Architecture Features:**
- Modular and scalable design (collection, processing, storage, and visualization layers).
- Real-time data processing pipeline.
- Fault-tolerant and easily extendable structure.

How Architecture Supports Non-Functional Requirements

1. **Performance – Process 10,000 log entries per second**

- Parallel ingestion from multiple Wazuh Agents.
- Optimized Logstash enrichment and time-based Elasticsearch indices.
- Horizontal scalability across ingestion tiers.

2. **Reliability – Recovery within 5 minutes after failure**

- Fault isolation between collection, processing, and storage tiers.
- Stateless components enable quick restart and auto-recovery.
- Health monitoring dashboards in Kibana.

3. **Efficiency – Memory usage under 1 GB**

- Lightweight event forwarding at the edge.
- Compact data indices in Elasticsearch.
- Minimal enrichment to reduce processing load.

How Architecture Supports Security Requirements

1. Broken Access Control

- Kibana is the only user-facing entry point.
- Role-based access: read-only for analysts, edit rights for engineers.
- Private network isolation for backend components (Wazuh, Logstash, Elasticsearch).

2. Input Manipulation Attack

- Early validation and normalization by Wazuh Agent and Manager.
- Strict schema enforcement—reject malformed logs.
- Alert triggers for suspicious log patterns or volume spikes.

3. Data Poisoning Attack

- Pre-processing and filtering of logs before storage.
- Separation of recent and archival data to limit contamination.
- Continuous performance reviews in Kibana to refine detection rules.

Thank You